

FLEA NEWS 24

Department of Entomology
Iowa State University
Ames, Iowa 50011 U.S.A.

May 1982

Compiled by Robert E. and Joanne H. Lewis

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Please find herewith:

- lists of additional literature for 1975 (list 14), 1976 (list 12), 1977 (list 10), 1978 (list 11), 1979 (list 10), 1980 (list 6), 1981 (list 3), and 1982 (list 1)
- additions, changes, and deletions to the mailing list

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CONGRATULATIONS - In a recent note from Dr. Lance A. Durden he writes: "On 1 May 1981, I successfully defended my PhD dissertation at London University, U.K., its title being Ectoparasites and arthropod associates of the eastern chipmunk, *Tamias striatus* (L.) in Davidson County, Tennessee, U.S.A. ... 140 pages."

Congratulations also go to Stephen A. Smith, who obtained the Master of Science degree from Ohio State University, Columbus, Ohio, USA in the summer of 1981. His thesis is entitled Studies on the morphology and biology of the bat flea *Myodopsylla insignis* (Rothschild) (Siphonaptera: Ischnopsyllidae). 129 pages, 86 figs., 4 tables.

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It is with deep regret that we have learned of the death of Dr. JUDSON LINSLEY GRESSITT and his wife, MARGARET, in an airplane crash in South China on 26 April 1982. Dr. Gressitt was associated with the Department of Entomology, Bernice P. Bishop Museum, Honolulu, Hawaii, and played a major role in founding the journals Pacific Insects and the Journal of Medical Entomology. An obituary and bibliography are planned for one or both of these serials.

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MAILING LIST

ADDITIONS:

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The British Library, Science Reference Library (A), Periodicals Recording Section, Room 1202, 25 Southampton Buildings, Chancery Lane, London WC2A 1AW, ENGLAND

Fan Peifang, Department of Parasitology, Shanghai Agricultural College, Qibao, Shanghai, PEOPLE'S REPUBLIC OF CHINA

Iowa State University Library, Jean Cook, Serials Department, Iowa State University, Ames, Iowa 50011 USA

Library of the Dutch Entomological Society, Entomologisch Museum, Plantage Mid-
denlaan 64, 1018 DH Amsterdam, NETHERLANDS

Valentina Filippovna Sapegina, Biological Institute, Siberian Branch, Academy of
Sciences of the USSR, 11 Frunze Street, Novosibirsk - 91, 630091 USSR

Mr. Stephen A. Smith, Entomology, 103 Botany & Zoology, 1735 Neil Avenue,
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CHANGES:

Mr. C. R. Baird, University of Idaho, SW Idaho Research & Extension Center, Rt. 2,
Box 2126, Parma, Idaho 83660 USA

Dr. Wolfgang H. O. Dorow, Schumannstrasse 9, 6000 Frankfurt am Main 1, WEST GERMANY

DELETIONS:

Mr. K. Baker

Miss E. van den Broek

Dr. L. Kartman

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RECENTLY PUBLISHED - The Ecology of Ectoparasitic Insects by Adrian G.
Marshall, 1981 (ISBN 0-12-474080-4), Academic Press Inc. (London) Ltd., i-xvi,
459 pp., £33.00/\$69.50.

This book is required reading for anyone interested in ectoparasitic insects. In the first 11 chapters the author deals with specific aspects of the ectoparasitic mode of life from a comparative point of view. Chapter 1 discusses ectoparasitism in its various forms as it occurs in the Hexapoda, and includes 11 tables dealing with the classification of ectoparasites and their avian and mammalian hosts. Chapter 2 contains a brief treatment of the environment in which ectoparasites spend their lives and from which they derive their food. Chapter 3 treats field and laboratory methods, including sampling techniques, marking for recapture, preservation, and laboratory maintenance. Chapter 4 is a detailed discussion of the life cycles in ectoparasitic insects, treating such topics as mating and oviposition, immature and adult stages, and causes of parasite mortality. Chapter 5 discusses food and feeding activity, modifications in mouthparts, food consumption, and the effects of differing diets. Chapter 6 deals with how ectoparasites locate their hosts, and both active and passive transfer between hosts. Chapter 7 summarizes what is known about host associations among ectoparasites, and treats host specificity, host resistance, and how ectoparasites have been used to deduce host phylogeny. Chapter 8 discusses the morphological and behavioural adaptations which allow the ectoparasite to remain on or near the host. Chapter 9 considers the interrelationships of ectoparasites with other organisms including their own parasites, competition among themselves, and other aspects contributing to their survival. Chapter 10 treats the population dynamics of ectoparasites, and Chapter 11 their effect upon the health of their host. Chapter 12 contains a brief summary of the author's conclusions. In addition to the list of species mentioned in the text and an index, there are 53 pages of bibliography.

To write a book of this type was an ambitious undertaking and the results are truly impressive, both in quantity and in quality. The format is pleasing to the eye and the illustrations are appropriate and well executed. In the

words of Miriam Rothschild who wrote the forward, "One word occurred to me when I finished reading the book: indispensable". ---R.E.L.

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RECENTLY PUBLISHED - Biology of Insect Eggs by Howard E. Hinton, 1981 (ISBN 0-08-021539-4) Pergamon Press Ltd., Vol. I, i-xxiv, 1-473; Vol. II, i-xviii, 475-778; Vol. III, i-xvii, 779-1125. U.S. \$400.00.

The publication of this long-awaited opus adds a new dimension to the study of insects. Fleas are only mentioned briefly on pages 10 (Aphaniptera), 26 (Siphonaptera, Pulicidae, Pulex irritans), and 176 (Xenopsylla), but the wealth of information concerning the other orders of insects makes these volumes an exceedingly valuable addition to the entomological literature. A detailed review of these volumes is beyond the purview of this newsletter, but following is a brief discussion of their content.

Volume I begins with an Introductory Note by Colin Mapes who was ultimately responsible for publication of the work, followed by an appreciation of the late H. E. Hinton by T. R. E. Southwood. Following these are an introduction and 11 chapters dealing with the following subjects: Number of Eggs, Oviposition, Respiratory Systems, Respiratory Efficiency of Egg Plastrons, Hydropyles and Water Relations, Colleterial Glands, Oothecal and Shell Proteins, Enemies of Eggs, Defensive Devices, Parental Care and Techniques. This volume also contains 155 photomicrographs of insect eggs, mostly taken on a scanning electron microscope.

Volume II contains a systematic treatment of the hexapod orders, dealing with such matters as parthenogenesis, oviposition, chorionic structure, micropyles, egg-bursters, diapause, and hatching. Where appropriate, other subjects are discussed and, in a few groups, keys to the eggs are provided. The volume concludes with an addendum listing papers on reproductive systems and micropylar apparatus, and a taxonomic list of species illustrated in volumes I and II.

Volume III contains an exhaustive bibliography of 220 pages, and indexes to species, authors, and subjects. It terminates with a bibliography of Hinton's publications from 1930 to 1977.

No doubt this work will remain the standard reference for decades to come. and although the price is prohibitive for individuals, it is a must for institutional libraries. ---R.E.L.

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NOTES

OUT OF BUSINESS - In Flea News No. 5 (V-1975) it was noted that copies of the Catalogue could be obtained in North America from the British Book Center in New York City. Subsequently the Center has gone out of business.

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By necessity we depend on a number of sources when translating or transliterating names of geographic localities. However, whenever possible, we follow the spelling given in The Times Atlas of the World, Mid-Century Edition, volumes I-V, 1955-1959, edited by John Bartholomew. In some cases alternative spellings are included in []. Corrections will be appreciated.

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NOMENCLATURAL COMMENTS - In 1980 a number of papers dealing with Siphonaptera were published in Acta Academiae Medicinae Guiyang 5(2). Through the kindness of Dr. Li Kuei-chen we are informed that these papers were published in December of that year. Some of the names mentioned in these papers are listed as in press. We have not yet seen descriptions so are unaware of their actual year of publication. However, there are a few nomenclatural problems which have been cleared up with Dr. Li's assistance. All of the papers have been listed in the 1980 section of Flea News 23 and will not be completely cited again here.

In the Li paper on Macrostylophora, the species luchunensis is variously cited as luchuensis Huang (in press), luchuensis Huang (in press), and Luchunensis Huang (in press). The actual description of this taxon appears in a later paper in this series and the name is spelled luchunensis. The description was republished in another journal in 1981 but the name should be cited as: Macrostylophora luchunensis Huang, 1980.

In the paper by Li and Huang, Neopsylla Guizhowensis Liao, 1974 should read: Neopsylla kweichowensis Liao, 1974.

Also in this paper Ctenophthalmus eothenomus sp.n., C. brevijiciens sp.n., and Typhlomyopsyllus n.gen. cavaticus n.sp. were described, as well as mention being made of Paraceras laxisinus He, Hsieh & Li in press. All but the latter name were listed in the English portion of the paper, and their descriptions subsequently republished in another journal in 1981. The status of these names is as follows:

1. Ctenophthalmus eothenomus Li & Huang, 1980

2. Ctenophthalmus brevijiciens

Although this name is cited both at the beginning of the description and in the caption of figure 3, it is listed as C. breviprojiciens sp. nov. in the English portion of the paper and three times in the subsequent republication. Since the original name does not appear to be based on any word of classical origin but the latter name does, it is suggested that the proper citation be Ctenophthalmus breviprojiciens Li & Huang, 1980.

3. Typhlomyopsyllus cavaticus Li & Huang, 1980

This name was spelled as shown above in the Chinese portion of the paper, but spelled Typhlomyopsylla in the English summary. In the 1981 republication, the authors reverted to the original spelling. The name should be cited as listed above.

4. Paraceras laxisinus He, Hsieh & Li sp.nov.

This species was described in a later paper in this series written by a different combination of authors. It should be cited: Paraceras laxisinus Xie, He & Li, 1980 (nec He, Hsieh & Li, 1980).

The status of some other names mentioned in this series is still unresolved and it is hoped that receipt of additional reprints will clarify their positions.

A. PREFATORY NOTE TO THE SPECIES LIST - In Flea News 23 we indicated our intention of assigning species numbers to those taxa named in 1980 in this issue of Flea News. Some recent papers suggest that there are additional species that may have been described in 1980 that we do not yet know about. In order to avoid confusion in later listings, we are following our practice of including only a provisional list of new names published since January 1980, and urge our readers to call any omissions to our attention as soon as possible.

1980

Macrostylophora paoshanensis Li & Yan
Ceratophyllus nanshanensis Tsai, Pan & Liu
Cratynius (Angustus) yunnanus Li, Xie & Liao
Mesopsylla hebes clara Smit
Hystrihopsylla rotundisinuata Li & Hsieh
Megathoracipsylla pentagonia Liu, Liu & Zhang
Genoneopsylla angustidigita Wu, Wu & Tsai
Genoneopsylla claviprocera Hsieh & Wu
Genoneopsylla bisinuata Liu, Chang & Liu
Vermipsylla qilianensis Wu, Tsai & Liu
Callopsylla kazbegiensis Goncharov
Geusibia aprominus Liu, Cai & Wu
Neopsylla fimbriata Li & Hsieh
Neopsylla elesina Li
Neopsylla megaloba Li
Neopsylla schismatosa Li
Ctenophthalmus eothenomus Li & Huang
Ctenophthalmus breviprojiciens Li & Huang
Typhlomyopsyllus cavaticus Li & Huang
Amphalius spirataenius diqingensis Li, Xie & Yang
Amphalius spirataenius qinghaiensis Li, Xie & Yang
Macrostylophora luchunensis Huang
Paraceras laxisinus Xie, He & Li
Paraceras menetus Xie, Chen & Li

1981

Nearctopsylla pfitzeri Benton
Amphipsylla kulkarnii Lewis
Vermipsylla ibexa Zhang & Yu
Araeopsylla faini Beaucournu
Ctenophthalmus (Ctenophthalmus) bisoctodentatus suciuae Beaucournu & Grulich
Macrostylophora abazhouensis Liu, Liu & Zhai
Amphalius spirataenius badongensis Ji, Chen & Wang
Amphalius clarus kunlunensis Yu & Wang
Ischnopsyllus (Hexactenopsylla) shansiensis Liu, Hsing & Chen
Nosopsyllus (Nosopsyllus) chayuensis Wang & Liu
Ctenophthalmus (Idiostenophthalmus) hopkinsi rahmi Beaucournu
Ctenophthalmus (Idiostenophthalmus) elblae Beaucournu

* * *

An asterisk (*) preceding a citation indicates we have seen neither the original publication nor verified it in another source. Anything appearing in brackets ([]) after a citation indicates an annotation, e.g., the name of a taxon described.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1975 (list 14)

- *Pissarska, M. & D. Mitev - Aphaniptera in small mammal nests in the western Rodop Planina [Rhodope Mts.]. Plovdivski Univ. "Paisij Hildenarskii" Nauchni Trudove Biol., Plovdiv 13(4): 189-203, 1 fig.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1976 (list 12)

- Sapegina, V. F. - Peculiarities of the distribution of fleas from small mammals and birds in the southern taiga of the Pri-Irtish. Parazitologiya 10(5): 397-400.
- Sapegina, V. F. & Yu. S. Ravkin - To knowledge of the flea fauna of the northern taiga of Pri-Obya. VII Symposium "Biological Problems in the North. Invertebrate Zoology, Parasitology, Physiology and Animal Biochemistry." Petrozavodsk.
- Verhulst, A. - Tunga penetrans (Sarcopsylla penetrans) as a cause of agalactia in sows in the Republic of Zaire. Vet. Rec. 98(18): 384.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1977 (list 10)

- *Bailly-Choumara, H. - Bibliographie entomologique marocaine. Retrospective de 1870 à 1975 inclus. Documents de l'Institut scient., Rabat, No. 1, 1-216.
- *Bogdanov, I. I. & V. A. Ilin - Zoologico-parasitological characteristics of an elevated zooanthropological nidus in eastern Taymyr. Problems in epidemiology and prophylaxis in an elevated nidus of disease in Zapolyare. Omsk, 11-23.
- Viljoen, S. - Ectoparasites of the bush squirrel, Paraxerus cepapi cepapi, in the Transvaal (Rodentia: Sciuridae). Zool. Africana 12(2): 498-500.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1978 (list 11)

- *Igolkina, N. I. - The ectoparasite complex in small mammals in the southeastern part of western Siberia. Tomsk.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1979 (list 10)

- Anonymous - In memory of Valentina Alexandrovna Bibikova. Med. Parazit., Moskva 48(4): 89-90.
- Austin, V. H. - Chapter 17. The skin. In: Canine Medicine, 4th ed. Vol. II. E. J. Catcott, Ed. American Veterinary Publications, Santa Barbara, Calif., USA, 1207-1247 [Siph: 1238-1240]
- Barlet, J. - Questions a propos des muscles trochantero-notaux des insectes. Bull. Ann. Soc. r. belge Ent. 115(1-3): 93-111. [Siph: 99, 103]
- ✓ Cerny, V. - Review of Tiflov, V. E., O. I. Scalon & B. A. Rostigayev, Opre-delitel blokh Kavkaza. [Key for determination of fleas of the Caucasus.] Publ. House Stavropolsk. knizhnoe izd. Stavropol' 1977, 278 pp., 384 figs. Folia parasit. (Praha) 26(1): 95.

(1979-1980)

- Cordero del Campillo, M. & F. A. Rojo Vazquez - La parasitologia veterinaria en las obras de albeyteria y afines. II. Fray Miguel Agustin (s. XVII-XVIII). Rev/ta iber. Parasitol. 39: 209-221. [Siph: 218-219]
- Emel'yanov, P. F., L. V. Bryukhanova, N. V. Glushko, N. P. Kalmykova, A. G. Reitblat & L. A. Tiflova - Some aspects of the ecology of the little suslik in isolated colonies in the Stavropol' Territory. Ekologiya 10(3): 69-74. [English translation of Ekologiya 10(3): 229-234]
- Green, R. H. - The little pigmy possum, *Cercartetus lepidus*, in Tasmania. Rev. Queen Vict. Mus. 68: 1-12. [Siph: 10]
- Green, R. H. - A survey of the vertebrate fauna of a heathland area between Round Hill and the Gardens, north-east Tasmania. Rec. Queen Vict. Mus. 67: 1-11. [Siph: 11]
- Jacobs, D. E. - Chapter 8. Man and his pets. In: Parasites and western man, R. J. Donaldson, Ed. University Park Press, Baltimore, v-x, 1-220. [Siph: 184-186]
- Maunder, J. W. - Chapter 2. Insects (Hexapoda). In: Parasites and western man. R. J. Donaldson, Ed. University Park Press, Baltimore, v-x, 1-220. [Siph: 21-26]
- Nilsson, A. & L. Lundqvist - Interspecific relations in small mammal ectoparasites. In: Recent advances in acarology, Vol. 1. J. G. Rodriguez, Ed. Academic Press, 451-456, 4 figs.
- Suzuki, S. & S. Hotta - Anti plague antibodies against *Yersinia pestis* Fraction-I antigen in serum from rodents either experimentally infected or captured in harbor areas of Japan, 1971-1977. Microbiol. Immunol. 23(12): 1157-1168.
- Thorson, T. E. - Flea bite allergic dermatitis in a red fox (*Vulpes fulva*). J. zoo Anim. Med. 10(1): 30.
- Vazquez, G. L. & C. S. Zaragoza - Tipos existentes en la coleccion entomologica del Instituto de Biologia, de la Universidad Nacional Autonoma de México. An. Inst. Biol. Univ. Nal. auton. México 50 (Zool.) no. 1: 575-632. [Siph: 590-591, 593, 595, 598, 600]
- Whitfield, P. J. - The biology of parasitism: an introduction to the study of associating organisms. Edward Arnold Publishers Ltd., London, v-x, 1-277. [Siph: 88, 152-153]
- Wink, M., C. Wink & D. Ristow - Parasitenbefall juveniler und adulter Gelbschnabelsturmtaucher (*Calonectris diomedea*). Bonn. zool. Beitr. 30(1-2): 217-219.
- Soldatkin, I. S. - Review of Rotshild, E. V., Spatial structure of natural focus of plague and methods of its study. Moscow State University, Moscow, 1978, 192 pp. Med. Parazit., Moskva 48(4): 80-81.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1980 (list 6)

- Anonymous - Human plague cases increasing slightly in U.S. J. amer. Med. Assoc. 244(16): 1764.
- Anonymous - Human plague in 1979. Wkly. epidemiol. Rec. 55(32): 241-244.

(1980)

- Anonymous - January's cover. Antenna 4(2): 37. [January's cover picture was scanning electron micrograph of adhesive antennal setae of a ♂ *Hystrichopsylla talpae* (Curtis), photo by Nigel Stork]
- *Ata-ur-Rahim, M. - Medical literature on Pakistan from 1840-1947: a bibliography. J. Pakistan med. Assoc. 30(11): 257-258; 30(12): 281-282 conclusion.
- Baillie, H. D. & J. C. Wood - Pyrethroids, their use in the control of animal ectoparasites. In: Developments in animal and veterinary sciences, Volume 6. "Trends in veterinary pharmacology and toxicology". Proc. 1st European Congr., 25-28 Sept 1979, Zeist, 256-260. [Siph: 259]
- Barriga, O. O. - The immunology of parasitic infections. A handbook for physicians, veterinarians, and biologists. University Park Press, Baltimore, v-x, 1-354. [Siph: 284-285, 287, 296-297]
- Beaucournu, J. C. - Les ectoparasites du lapin de garenne, *Oryctolagus cuniculus*: apports a son histoire. Bull. mens. off. Nation. Chasse, No. Sp. Scien. Techn. (Dec.): 23-35.
- Beaucournu, J. C., H. Launay, H. Bailly-Choumara & M. Ramdani - Notes sur quelques *Hystrichopsyllidae* (Siphonaptera) du Maroc. Bull. Inst. Scientif. Rabat 1979-80(4): 29-40, figs. 1-4.
- Bennett, G. W. & E. S. Runstrom - Efficacy of new insecticide formulations in urban pest control. Pest Control 1980(Dec.): 19-22, 24.
- Bres, P. L. - Could the great "plagues" recur? World Health 1980(Nov.): 7-9.
- Carr, S. H. - Clinical observations on the topical use of fenthion. Canine Pract. 7(4): 69-70, 72.
- *Castillo, J. M. - Pathogens of Siphonaptera (fleas). Bull. Wrld. Hlth. Org., Suppl. to 58 [Bibliography on pathogens of medically important arthropods, D. W. Roberts & J. M. Castillo, eds.]: 141-149.
- *Castillo, J. M. - Pathogen-host lists. Bull. Wrld. Hlth. Org., Suppl. to 58 [Bibliography of pathogens of medically important arthropods, D. W. Roberts & J. M. Castillo, eds.]: 169-197.
- de Carneri, I. - Da *Leishmania major* a *Tunga penetrans*: parassiti esotici arrivano in Italia. Parassitologia 22(3): 300-301.
- *Eckert, E. A. - Seasonality of plague in early modern Europe: Swiss epidemic of 1628-1630. Rev. infect. Dis. 2(6): 952-959.
- Fallis, A. M. - Arthropods as pests and vectors of disease. Vet. Parasitol. 6(1-3): 47-73. [Siph: 62-63]
- Faubleé, V. - Les relations entre les ectoparasites permanents obligatoires et leur hôtes. Synthèse bibliographique. Rec. Méd. vét. 156(3): 225-236.
- Frazier, C. A. & F. K. Brown - Insects and allergy and what to do about them. University of Oklahoma Press, Norman, vii-xiv, 1-272. [Siph: 7, 10, 13, 47, 170-178]
- Gadzhiyev, A. T., Z. A. Mustafayeva & T. A. Dubovchenko - Ectoparasite fauna of *Hirundo rustica* (Passeriformes, Hirundinidae) and their nests in Azerbaydhan. Izv. Akad. Nauk azerbaydz. SSR (Biol.) 1980(6): 76-82.

(1980)

- Green, D. B. - The fauna of a Lancashire deep-pit poultry house. Entomol. Month. Mag. 116: 13-15.
- Hancox, M. - Parasites and infectious diseases of the Eurasian badger (*Meles meles* L.): a review. Mamm. Rev. 10(4): 151-162.
- Harrison, D. N., W. Laird, D. M. Robinson & D. C. Cavanaugh - Commonality of a virulence factor among *Yersinia* species. J. infect. Dis. 141(3): 413.
- *Hartwig, E. K. - Plague surveillance. Proc. 3rd Entomol. Congr., Ent. Soc. S. Afr., Pretoria, 16-18 Sept.: 44.
- *Heckly, R. J. & H. Blank - Virulence and variability of *Yersinia pestis* 25 years after lyophilization. Appl. environm. Microbiol. 39: 541-543.
- Hudson, B. W., K. Wolff & T. Butler - The use of solid-phase radioimmunoassay techniques for serodiagnosis of human plague infection. Bull. Pan Amer. Hlth. Org. 14(3): 244-250.
- *Humke, R., W. Bonin & W. Thier - Ragadan, a new agent for control of ectoparasites (*Heptenophos*). Blauen Hefte Tierarzt 62: 82-87.
- Laird, W. J. & D. C. Cavanaugh - Correlation of autoagglutination and virulence of *Yersinia*. J. clin. Microbiol. 11(4): 430-432.
- Launay, H. - Approche d'une prophylaxie de la myxomatose: Ecologie des puces du lapin de Garenne. Bull. mens. off. Nation. Chasse, No. Sp. Scien. Techn. (Dec.): 213-241.
- Liu, C.-y., L.-y. Cai & W.-z. Wu - Description and discussion on a new species of *Geusibia* from Qinghai Province, China. Entomotaxonomia 2(3): 173-176, figs. 1-3. [*Geusibia aprominus* Liu, Cai & Wu]
- Liu, C.-y., Y. Liu & J. Zhang - Description and discussion of a new genus and species of *Ceratophyllidae* from Shaanxi-Province, China. Entomotaxonomia 2(2): 135-139, figs. 1-5. [*Megathoracipsylla pentagonia* Liu, Liu & Zhang]
- Liu, C.-y., H. Wu & G. Shao - On *Nycteridopsylla* Oudemans, 1906 in China (Siphonaptera: *Ischnopsyllidae*). Entomotaxonomia 2(4): 291-296, figs. 1-4.
- Miller, W. H. - Shampoos in veterinary dermatology. Calif. Vet. 34(11): 9-11.
- Sagdieva, P. D. - Ectoparasitic mites and ticks (*Parasitiformes*) and fleas (*Siphonaptera*) of small rodents in the forests of middle Sykhote-Alyn. Soobšč. Akad. Nauk gruzinskoj SSR, Tbilisi 100(2): 492-496.
- *Sapegina, V. F., Yu. S. Ravkin & I. V. Lukyanova - Fleas of the forest zone of western Siberia. In: Problems in zoogeography and faunal history. Novosibirsk: Nauka.
- *Sapegina, V. F., L. G. Vartapetov & S. M. Ts'bulin - The fleas of small mammals in the northern and central taiga of Pri-Obya. Trud. biol. Inst. Akad. Nauk SSSR (Siberian Branch) Novosibirsk 42: 232-235. [Parasitic insects and ticks of Siberia, M. S. Davydova, Ed.]
- *Sapegina, V. F., B. S. Yudin & G. V. Dudareva - Material on the biology of fleas from the Taymyr and Gdanskogo Peninsulas. Trud. biol. Inst. Akad. Nauk SSSR (Siberian Branch) Novosibirsk 42: 225-231. [Parasitic insects and ticks of Siberia, M. S. Davydova, Ed.]
- *Soci, K. I. & H. J. Van der Kaay - Tungiasis: a case report. Acta Leidensia 48: 43-46.

(1980-1981)

- Soriguer, R. C. - Ciclo anual de parasitismo por pulgas y garrapatas en el conejo de campo (*Oryctolagus cuniculus* L.) en Andalucía Occidental, España. Rev/ta iber. Parasitol. 40(4): 539-550.
- Voronov, A. G. - Advances in the study of plague foci. Review of E. V. Rotshild, Spatial structure of natural plague foci and methods of its study. Moscow University Press, 1978, 192 pp. Bull. Moskovskoe Obshchest. ispyt. prirody 85(4): 114-116.
- Waage, J. - Curse of the vampire: the evolution of blood-sucking insects. Antenna 4(4): 112-116.
- Wharton, R. H. & K. R. Norris - Control of parasitic arthropods. Vet. Parasitol. 6(1-3): 135-164. [Siph: 141]
- *Wirtz, R. A. & L. G. Zárate - Alergias causadas por insectos. Salud publ. México, época 5, 22(6): 595-599.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1981 (list 3)

- Ade-Serrano, M. & G. Chuks Ejezie - Prevalence of tungiasis in Oto-Ijanikin village, Badagry, Lagos State, Nigeria. Ann. trop. Med. Parasitol. 75(4): 471-472.
- Amrine, J. W. - Chapter 16. Scanning Electron Microscopy. In: Principles of food analysis for filth, decomposition and foreign matter. J. R. Gorham, Ed. FDA Tech. Bull. No. 1, 2nd ed., 233-245. [Contains scanning electron photomicrographs of *Cediopsylla simplex*]
- Anonymous - Human plague in 1980. Wkly. epidemiol. Rec. 56: 273-275.
- Anonymous - Instructions for determining the susceptibility or resistance of fleas to insecticides. WHO/VBC/81.815: 1-6.
- Anonymous - Purdue theme hails professional specialization for profits in the 80's. [Conference report] Pest Control 49(4): 16-18, 21-22, 24 [Siph: 22]
- Anonymous - Toxic flea repellent reported. J. amer. vet. Med. Assoc. 179(3): 212.
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It will be understood that - although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in the USSR or China, are in Russian or Chinese (but often with summaries in English).



PCHŁA

PCHŁY, przyjaciele, ukochana,
Ci co kochają, okrutni są dla nas.
Ich miłowanie do krwi dręczy.
Nie są szczęśliwi oblubieńcy.

Fleas, dear friends, are cruel
To those who are in love,
Because of their great love of blood.
They are terrible to us who make love.
Therefore, lovers cannot be happy.